

Justin Blackwood

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EDUCATION

University of Florida (GPA 3.66)

B.S. in Computer Science

Gainesville, Florida

Aug. 2024 – Anticipated May 2027

EXPERIENCE

Intro to Programming Teaching Assistant

University of Florida

August 2025 – Present

Gainesville, Florida

- Leading two weekly Programming I lab sessions of **60** students, guiding them through foundational coding concepts and **Python** based hands-on Labs.
- Held weekly office hours to provide individualized support and debugging code for **20+** first-year CS student.
- Contributed feedback and insights in weekly team meetings with faculty leading to course restructuring that resulted in a **17%** increase in lab averages.

Testing & QA Coordinator

University of Florida

August 2025

Gainesville, Florida

- Coordinated a team of **10** testers, designing and facilitating structured testing sessions to identify defects, usability issues, and performance gaps.
- Managed bug tracking and resolution, working directly with testers and developers to confirm bug fixes successfully identifying and resolving **7** critical bugs.
- Delivered a production-ready Discord bot now in active use by **25** users handling seasonal transactions and notifications.

PROJECTS

Probabilistic Race Simulation System | *Flask, React.js, Git*

January 2026 - Present

- Collaborating on the development of a Python-based race simulation platform, implementing probabilistic models to generate outcomes, rankings, and performance distributions.
- Designing backend systems to support a planned web-based frontend, including participant state management, and leaderboard computation.
- Actively translating spreadsheet-defined rules into modular, maintainable backend code, with ongoing iteration and validation during development.
- Implementing persistent data models for users, events, and results to support repeatable simulations and historical analysis as the system evolves.

Real-Time Networked Chess | *C++*

December 2025 - Present

- Developing a real-time simulation variant of chess with concurrent player actions.
- Implementing a LAN networking layer to transmit player commands and synchronize game state.
- Designing a deterministic update loop to ensure consistent outcomes across networked clients.

Draft League Bot | *Python, Json, Github*

July 2025 - Present

- Designed and deployed a custom bot to automate drafting, point tracking, and rule enforcement, reducing admin work by **80%**.
- Collaborated with a contributing developer via GitHub to review, integrate, and test new features, while maintaining consistency with existing systems and league rules.
- Communicated with league Admin through design and testing process to ensure all specifications were met to satisfactory levels.

TECHNICAL SKILLS

Languages: Python, C++, HTML, CSS

Developer Tools: Git/GitHub, Visual Studio Code, Linux, Cloudflare

Libraries: SFML, Discordpy, Pygame, socket

Technologies: JSON, Node.js, SQLite, React.js

Relevant courses: Data Structures and Algorithms, Computer organization, Computer Networking, Operating Systems